Channel Estimation for HF Skywave Massive MIMO-OFDM with Triple-Beam Based Channel Model

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***Abstract*—In this paper, we investigate channel estimation for high frequency (HF) skywave massive multiple-input multiple-output (MIMO) communications with orthogonal frequency division multiplexing (OFDM) modulation. We first introduce the concept of triple beams (TBs) in the space-frequency-time (SFT) domain and establish a TB based channel model using sampled triple steering vectors. With the established channel model, we then investigate the optimal channel estimation and pilot design for pilot segments. Specifically, we find the conditions that allow pilot reuse among multiple user terminals (UTs), which significantly reduces pilot overhead. To reduce the complexity, we are able to formulate the channel estimation as a sparse signal recovery problem due to the channel sparsity in the TB domain and then obtain the channel by the proposed constrained Bethe free energy minimization (CBFEM) based channel estimation algorithm. Simulation results demonstrate the superior performance of the proposed channel estimation approach.**

I. Introduction

High frequency (HF) communications, whose frequency range is usually from 3 MHz to 30 MHz, can provide world-wide coverage through skywave propagation. Compared with satellite communications, an alternative for global coverage, HF communications can be flexibly deployed with relatively low cost and are robust to jamming [1]. However, the low data transmission rate of HF communications makes it hard to compete with satellite communications.

In the past decade, massive multiple-input multiple-output (MIMO) has been widely studied for terrestrial cellular communications. It can significantly improve the spectrum and power efficiencies [2] and has been introduced into the HF skywave communications with orthogonal frequency division multiplexing (OFDM) modulation in [3] recently. It is demonstrated in [3] that massive MIMO can vastly improve the performance of HF skywave communications.

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The acquisition of the channel state information (CSI) is essential for massive MIMO to harvest the significant performance gain, which has been well investigated in terrestrial cellular massive MIMO systems [4]–[12]. The conventional orthogonal pilot signaling and channel estimation methods, such as least squares (LS) and minimum mean-squared error (MMSE) [4], are not suitable to the massive MIMO system due to the overwhelming pilot overhead and the forbidding computational complexity. By exploiting the constraint of the angle spread in the massive MIMO, the pilot reuse schemes have been proposed to reduce the pilot overhead, where user terminals (UTs) with non-overlapping angle domain channels are allowed to share the same pilot [5], [6]. On the other hand, due to the limited local scatters, the terrestrial massive MIMO channel exhibits sparsity, which enables channel estimation through sparse signal recovery [7]–[10]. Moreover, channel estimation approaches highly depend on the channel models. The spatial-beam based channel model has been widely used in [5], [8] for channel estimation. An accurate spatial-beam based channel model using finely sampled steering vectors in [11], [12] can improve channel estimation performance.

In this paper, we investigate channel estimation for HF skywave massive MIMO-OFDM communications. We introduce the concept of the triple beams (TBs) in space-frequency-time (SFT) domain and use it to derive a TB based channel model. Based on the TB based channel model, we find the conditions to minimize the normalized mean-squared error (NMSE) of optimal estimates for channels at pilot segments. Specifically, we show that UTs with overlapping TB domain channels should be allocated pilot sequences with different phase shift factors while UTs with non-overlapping TB domain channels can reuse the same pilot sequence. Subsequently, we develop a constrained Bethe free energy minimization (CBFEM) based channel estimation algorithm to reduce the complexity.

II. Channel Model

A. System Configuration

We consider a HF skywave massive MIMO-OFDM system. The Base station (BS) is equipped with a uniform linear array

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(ULA) with M antennas and serves U single-antenna UTs. Let N_{c} denote the number of subcarriers, N_{g} denote the length of the cyclic prefix (CP), Δf denote the subcarrier spacing, and $T_{\mathrm{s}}=\frac{1}{N_{\mathrm{c}}\Delta f}$ denote the sampling interval. We assume that N_{v} valid subcarriers are used to transmit data and pilots with the index set $\mathcal{K}=\{k_0,k_1,\cdots,k_{N_{\mathrm{v}}-1}\}$.

The HF skywave massive MIMO-OFDM system operates in time-division duplex (TDD) mode with frame structure shown in Fig. 1. Each frame consists of N_{F} timeslots and each timeslot consists of N_{S} OFDM symbols, thus the number of OFDM symbols in each frame is $N=N_{\mathrm{F}}N_{\mathrm{S}}$. In each timeslot, the n_{p}-th OFDM symbol is the pilot used for uplink training and the other OFDM symbols are used for data transmission.

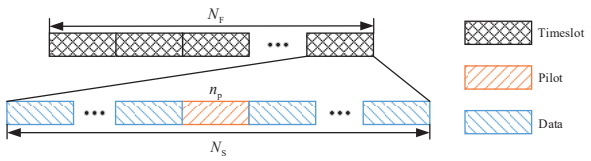


Fig. 1. The frame structure.

In HF skywave communications, the carrier frequency f_{c} varies with ionospheric conditions at different times. Hence, we set inter-antenna spacing d according to the highest system operating frequency f_{o}, that is, $d=\lambda_{\mathrm{o}}/2$, where $\lambda_{\mathrm{o}}=c/f_{\mathrm{o}}$ is the wavelength, and c is the speed of light. This is different from the traditional massive MIMO system, where the antenna spacing is usually half the wavelength of the carrier.

B. Triple-Beam Based Channel Model

Let $x_{u,n,k}$ denote the transmitted data of UT u on the n-th OFDM symbol at the k-th subcarrier, where $n\in\{0,1,\cdots,N-1\}$ and $k\in\mathcal{K}$. After OFDM modulation, the analog baseband signal on the n-th OFDM symbol transmitted by UT u with the CP can be expressed as

$$\bar{x}_{u,n}(t)=\sum_{k=k_0}^{k_{N_{\mathrm{v}}-1}}x_{u,n,k}e^{-\bar{\jmath}2\pi k\Delta f t},\ -N_{\mathrm{g}}T_{\mathrm{s}}\le t-nT_{\mathrm{sym}}<N_{\mathrm{c}}T_{\mathrm{s}},\tag{1}$$

where $T_{\mathrm{sym}}=(N_{\mathrm{c}}+N_{\mathrm{g}})T_{\mathrm{s}}$ is the time duration of an OFDM symbol including CP. At the BS, the received analog baseband signal on the n-th OFDM symbol at the m-th antenna is

$$\bar{y}_{m,n}(t)=\sum_{u=0}^{U-1}\int_{-\infty}^{\infty}\bar{h}_{u,m}(t,\tau)\,\bar{x}_{u,n}(t-\tau)\,d\tau+\bar{z}_{m,n}(t),\tag{2}$$

where $m\in\{0,1,\cdots,M-1\}$, $\bar{z}_{m,n}(t)$ is the additive white Gaussian noise and $\bar{h}_{u,m}(t,\tau)$ is the time-varying channel impulse response between UT u and the m-th antenna of the BS. The channel impulse response can be expressed as

$$\bar{h}_{u,m}(t,\tau)=\sum_{p=0}^{P_u-1}\gamma_{u,p}e^{\bar{\jmath}2\pi\nu_{u,p}t}e^{-\bar{\jmath}2\pi f_{\mathrm{c}}m\Delta\tau\Omega_{u,p}}\delta(\tau-\tau_{u,p}-m\Delta\tau\Omega_{u,p}),\tag{3}$$

where P_u is the number of paths between UT u and the BS, $\gamma_{u,p}$, $\nu_{u,p}$, and $\Omega_{u,p}$ is the complex-valued gain, the Doppler frequency, and the directional cosine of the p-th path of UT u, respectively, $\tau_{u,p}$ is the p-th propagation path delay between UT u and the first antenna of the BS, and $\Delta\tau=d/c$. In (3), the complex-valued gain can be expressed as $\gamma_{u,p}=\beta_{u,p}e^{\bar{\jmath}\varphi_{u,p}}$, where $\beta_{u,p}$ and $\varphi_{u,p}$ are the gain and initial phase, respectively, and $\varphi_{u,p}$ is uniformly distributed over $[0,2\pi)$. The directional cosine is defined as $\Omega_{u,p}\triangleq\sin\theta_{u,p}^{\mathrm{az}}\cos\theta_{u,p}^{\mathrm{el}}$, where $\theta_{u,p}^{\mathrm{az}}$ and $\theta_{u,p}^{\mathrm{el}}$ are the azimuth angle of arrival (AoA) and elevation AoA, respectively. Note that the propagation delay across the antenna array (i.e., $m\Delta\tau\Omega_{u,p}$ in the delta function) is considered due to the spatial-wideband effect caused by the equipped large-scale antenna array and wider bandwidth compared with traditional HF communications [3], [13].

We assume that the channel state keeps constant within an OFDM symbol and varies symbol by symbol due to the Doppler effect. After the OFDM demodulation, the received data at the k-th subcarrier on the n-th OFDM symbol at the m-th antenna is given by

$$y_{m,n,k}=\sum_{u=0}^{U-1}h_{u,m,n,k}^{\mathrm{SFT}}x_{u,n,k}+z_{m,n,k},\tag{4}$$

where $z_{m,n,k}$ is the noise with distribution $\mathcal{CN}\left(z_{m,n,k};0,\sigma_{\mathrm{z}}^2\right)$, and $h_{u,m,n,k}^{\mathrm{SFT}}$ is the channel frequency response at the k-th subcarrier on the n-th OFDM symbol between UT u and the m-th antenna at the BS, which is given by

$$h_{u,m,n,k}^{\mathrm{SFT}}=\sum_{p=0}^{P_u-1}\gamma_{u,p}e^{\bar{\jmath}2\pi\nu_{u,p}nT_{\mathrm{sym}}}e^{-\bar{\jmath}2\pi f_{\mathrm{c}}m\Delta\tau\Omega_{u,p}}\times e^{-\bar{\jmath}2\pi k\Delta f\tau_{u,p}}e^{-\bar{\jmath}2\pi k\Delta f m\Delta\tau\Omega_{u,p}}.\tag{5}$$

We consider the whole space-frequency domain channel over N OFDM symbols between UT u and the BS, which is referred to as the space-frequence-time (SFT) domain channel vector, $\mathbf{h}_u^{\mathrm{SFT}}$, with element $h_{u,m,n,k}^{\mathrm{SFT}}$ of index $(nMN_{\mathrm{v}}+(k-k_0)M+m)$. We denote

$$\mathbf{v}_k(\Omega)\triangleq\left[1,e^{-\bar{\jmath}2\pi(f_{\mathrm{c}}+k\Delta f)\Delta\tau\Omega},\cdots,e^{-\bar{\jmath}2\pi(f_{\mathrm{c}}+k\Delta f)(M-1)\Delta\tau\Omega}\right]^{\mathrm{T}},$$

$$\mathbf{u}(\tau)\triangleq\left[e^{-\bar{\jmath}2\pi k_0\Delta f\tau},e^{-\bar{\jmath}2\pi k_1\Delta f\tau},\cdots,e^{-\bar{\jmath}2\pi k_{N_{\mathrm{v}}-1}\Delta f\tau}\right]^{\mathrm{T}},$$

$$\mathbf{d}(\nu)\triangleq\left[1,e^{\bar{\jmath}2\pi\nu T_{\mathrm{sym}}},\cdots,e^{\bar{\jmath}2\pi\nu(N-1)T_{\mathrm{sym}}}\right]^{\mathrm{T}},$$

as steering vectors in the space domain, the frequency domain, and the time domain, pointing towards directional cosine Ω, delay τ, and Doppler frequency ν, respectively. Note that the steering vectors in the space domain are different for different subcarriers due to the spatial-wideband effect. Denote

$$\mathbf{p}(\Omega,\tau,\nu)\triangleq\mathbf{d}(\nu)\otimes(\mathbf{v}(\Omega)\circ(\mathbf{u}(\tau)\otimes\mathbf{e}_M))\in\mathbb{C}^{MN_{\mathrm{v}}N\times1},$$

where $\mathbf{e}_M\triangleq[1,1,\cdots,1]^{\mathrm{T}}\in\mathbb{C}^{M\times1}$, $\circ$ and $\otimes$ denote Hadamard and Kronecker product operators, respectively, and

$$\mathbf{v}(\Omega)\triangleq\left[\mathbf{v}_{k_0}(\Omega)^{\mathrm{T}},\mathbf{v}_{k_1}(\Omega)^{\mathrm{T}},\cdots,\mathbf{v}_{k_{N_{\mathrm{v}}-1}}(\Omega)^{\mathrm{T}}\right]^{\mathrm{T}}\in\mathbb{C}^{MN_{\mathrm{v}}\times1}.$$

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Therefore, the SFT domain channel can be expressed as

$$\mathbf{h}_u^{\mathrm{SFT}} = \sum_{p=0}^{P_u-1} \gamma_{u,p}\mathbf{p}\left(\Omega_{u,p},\tau_{u,p},\nu_{u,p}\right) \in \mathbb{C}^{MN_{\mathrm{v}}N\times 1}. \quad (6)$$

In this physical channel model, $\mathbf{p}\left(\Omega_{u,p},\tau_{u,p},\nu_{u,p}\right)$ represents a triple steering vector pointing towards the channel parameters $(\Omega_{u,p},\tau_{u,p},\nu_{u,p})$. Moreover, the complex-valued gain of a path, $\gamma_{u,p}$, is also the weight of the triple steering vector.

We assume that all UTs are synchronized. Note that the parameters $(\Omega_{u,p},\tau_{u,p},\nu_{u,p})$ of each path for each UT are limited within sets $\mathcal{B}_{\mathrm{an}} = \{\Omega\,|\Omega\in[-1,1)\}$, $\mathcal{B}_{\mathrm{de}} = \{\tau\,|\tau\in[0,\tau_{\max})\}$ and $\mathcal{B}_{\mathrm{do}} = \{\nu\,|\nu\in[-\nu_{\max},\nu_{\max})\}$, respectively, where $\tau_{\max}\triangleq\frac{N_{\mathrm{g}}}{N_{\mathrm{c}}\Delta f}$ is the maximum delay spread, and $\nu_{\max}\triangleq\frac{N_{\mathrm{d}}}{2NT_{\mathrm{sym}}}$ is the maximum Doppler [14], [15]. We uniformly divide these sets into multiple disjoint subsets as $\mathcal{B}_{\mathrm{an}}=\bigcup_{n_{\mathrm{an}}=0}^{N_{\mathrm{an}}-1}\Lambda_{n_{\mathrm{an}}}^{\mathrm{an}}$, $\mathcal{B}_{\mathrm{de}}=\bigcup_{n_{\mathrm{de}}=0}^{N_{\mathrm{de}}-1}\Lambda_{n_{\mathrm{de}}}^{\mathrm{de}}$,$\mathcal{B}_{\mathrm{do}}=\bigcup_{n_{\mathrm{do}}=0}^{N_{\mathrm{do}}-1}\Lambda_{n_{\mathrm{do}}}^{\mathrm{do}}$, where $\Lambda_{n_{\mathrm{an}}}^{\mathrm{an}}=\left[\frac{2n_{\mathrm{an}}-N_{\mathrm{an}}}{N_{\mathrm{an}}},\frac{2n_{\mathrm{an}}-N_{\mathrm{an}}+2}{N_{\mathrm{an}}}\right)$,$\Lambda_{n_{\mathrm{de}}}^{\mathrm{de}}=\left[\frac{N_\tau n_{\mathrm{de}}}{N_{\mathrm{de}}N_{\mathrm{v}}\Delta f},\frac{N_\tau(n_{\mathrm{de}}+1)}{N_{\mathrm{de}}N_{\mathrm{v}}\Delta f}\right)$, $\Lambda_{n_{\mathrm{do}}}^{\mathrm{do}} = \left[\frac{N_{\mathrm{d}}(n_{\mathrm{do}}-N_{\mathrm{do}}/2)}{N_{\mathrm{do}}NT_{\mathrm{sym}}},\frac{N_{\mathrm{d}}(n_{\mathrm{do}}-N_{\mathrm{do}}/2+1)}{N_{\mathrm{do}}NT_{\mathrm{sym}}}\right)$, $N_\tau\triangleq\frac{N_{\mathrm{v}}N_{\mathrm{g}}}{N_{\mathrm{c}}}$, and $\frac{2}{N_{\mathrm{an}}}$, $\frac{N_\tau}{N_{\mathrm{de}}N_{\mathrm{v}}\Delta f}$ and $\frac{N_{\mathrm{d}}}{N_{\mathrm{do}}NT_{\mathrm{sym}}}$ are the intervals in directional cosine, delay and Doppler frequency, which can be flexibly adjusted by properly selecting $N_{\mathrm{an}} = \left\lceil F_{\mathrm{an}}M\frac{f_{\mathrm{c}}}{f_{\mathrm{o}}}\right\rceil$, $N_{\mathrm{de}} = F_{\mathrm{de}}N_\tau$, and $N_{\mathrm{do}} = F_{\mathrm{do}}N_{\mathrm{d}}$, where F_{an}, F_{de}, and F_{do} are referred to as fine factors. Denote $\Gamma_u\triangleq\{(\Omega_{u,0},\tau_{u,0},\nu_{u,0}),\cdots,(\Omega_{u,P_u-1},\tau_{u,P_u-1},\nu_{u,P_u-1})\}$ as the path parameter set of UT u, and let $\Lambda_{n_{\mathrm{an}},n_{\mathrm{de}},n_{\mathrm{do}}}\triangleq\Lambda_{n_{\mathrm{an}}}^{\mathrm{an}}\times\Lambda_{n_{\mathrm{de}}}^{\mathrm{de}}\times\Lambda_{n_{\mathrm{do}}}^{\mathrm{do}}$, where $\times$ here denotes the Cartesian product. The SFT domain channel in (6) is rewritten as

$$\mathbf{h}_u^{\mathrm{SFT}}=\sum_{n_{\mathrm{an}}=0}^{N_{\mathrm{an}}-1}\sum_{n_{\mathrm{de}}=0}^{N_{\mathrm{de}}-1}\sum_{n_{\mathrm{do}}=0}^{N_{\mathrm{do}}-1}\sum_{(\Omega_{u,p},\tau_{u,p},\nu_{u,p})\in\Gamma_u\cap\Lambda_{n_{\mathrm{an}},n_{\mathrm{de}},n_{\mathrm{do}}}}\gamma_{u,p}\times\mathbf{p}\left(\Omega_{u,p},\tau_{u,p},\nu_{u,p}\right). \quad (7)$$

We approximate the triple steering vector $\mathbf{p}\left(\Omega_{u,p},\tau_{u,p},\nu_{u,p}\right)$ for $(\Omega_{u,p},\tau_{u,p},\nu_{u,p})\in\Gamma_u\cap\Lambda_{n_{\mathrm{an}},n_{\mathrm{de}},n_{\mathrm{do}}}$ as the sampled triple steering vector $\mathbf{p}\left(\Omega_{n_{\mathrm{an}}},\tau_{n_{\mathrm{de}}},\nu_{n_{\mathrm{do}}}\right)$, where $\Omega_{n_{\mathrm{an}}}=\frac{2n_{\mathrm{an}}-N_{\mathrm{an}}}{N_{\mathrm{an}}}$, $\tau_{n_{\mathrm{de}}}=\frac{N_\tau n_{\mathrm{de}}}{N_{\mathrm{de}}N_{\mathrm{v}}\Delta f}$, and $\nu_{n_{\mathrm{do}}}=\frac{N_{\mathrm{d}}(n_{\mathrm{do}}-N_{\mathrm{do}}/2)}{N_{\mathrm{do}}NT_{\mathrm{sym}}}$. This approximation tends to be accurate as the numbers of divided subsets N_{an}, N_{de} and N_{do} tend to be relatively large. We then define a matrix $\mathbf{P}$ that consists of sampled triple steering vectors and its $(n_{\mathrm{do}}N_{\mathrm{an}}N_{\mathrm{de}}+n_{\mathrm{de}}N_{\mathrm{an}}+n_{\mathrm{an}})$-th column is $\mathbf{p}\left(\Omega_{n_{\mathrm{an}}},\tau_{n_{\mathrm{de}}},\nu_{n_{\mathrm{do}}}\right)$, which represents the triple steering vector pointing towards the sampled directional cosine $\Omega_{n_{\mathrm{an}}}$, delay $\tau_{n_{\mathrm{de}}}$ and Doppler frequency $\nu_{n_{\mathrm{do}}}$. Then the SFT domain channel in (7) can be approximated as

$$\mathbf{h}_u^{\mathrm{SFT}} = \mathbf{P}\mathbf{h}_u^{\mathrm{TB}}, \quad (8)$$

where $\mathbf{h}_u^{\mathrm{TB}}\in\mathbb{C}^{N_{\mathrm{an}}N_{\mathrm{de}}N_{\mathrm{do}}\times 1}$ can be expressed as

$$\left[\mathbf{h}_u^{\mathrm{TB}}\right]_{n_{\mathrm{do}}N_{\mathrm{an}}N_{\mathrm{de}}+n_{\mathrm{de}}N_{\mathrm{an}}+n_{\mathrm{an}}}\triangleq\sum_{(\Omega_{u,p},\tau_{u,p},\nu_{u,p})\in\Gamma_u\cap\Lambda_{n_{\mathrm{an}},n_{\mathrm{de}},n_{\mathrm{do}}}}\gamma_{u,p}.$$

In the channel representation of (8), all the UTs share the same set of sampled triple steering vectors. Each sampled triple steering vector corresponds to a physical triple-beam in the SFT domain. Therefore, we refer to it as the *triple-beam (TB) based channel model*. $\mathbf{h}_u^{\mathrm{TB}}$ is the TB domain channel vector and $\mathbf{P}$ is referred to as the TB matrix. Furthermore, the SFT domain channel covariance matrix of UT u can be given by

$$\mathbf{R}_u^{\mathrm{SFT}} = \mathbb{E}\left\{\mathbf{h}_u^{\mathrm{SFT}}\left(\mathbf{h}_u^{\mathrm{SFT}}\right)^{\mathrm{H}}\right\} = \mathbf{P}\mathbf{R}_u^{\mathrm{TB}}\mathbf{P}\in\mathbb{C}^{MN_{\mathrm{v}}N\times MN_{\mathrm{v}}N},$$

where $\mathbf{R}_u^{\mathrm{TB}}\triangleq\mathbb{E}\left\{\mathbf{h}_u^{\mathrm{TB}}\left(\mathbf{h}_u^{\mathrm{TB}}\right)^{\mathrm{H}}\right\}\in\mathbb{C}^{N_{\mathrm{an}}N_{\mathrm{de}}N_{\mathrm{do}}\times N_{\mathrm{an}}N_{\mathrm{de}}N_{\mathrm{do}}}$ is the TB domain channel covariance matrix of UT u, also referred to as the TB domain statistical CSI, and $\mathbb{E}\{\cdot\}$ denotes the expectation. Note that the TB domain statistical CSI $\mathbf{R}_u^{\mathrm{TB}}$ varies slowly relative to the communication timescale [13], [15]. Thus we assume that the TB domain statistical CSI $\mathbf{R}_u^{\mathrm{TB}}$ is available at the BS in the rest of the paper. Moreover, we assume that the TB domain channel coefficients follow independent complex Gaussian distributions with zero mean and different variances. Hence, $\mathbf{R}_u^{\mathrm{TB}}$ is a diagonal matrix with the $(n_{\mathrm{do}}N_{\mathrm{an}}N_{\mathrm{de}}+n_{\mathrm{de}}N_{\mathrm{an}}+n_{\mathrm{an}})$-th diagonal element expressed as $\sum_{(\Omega_{u,p},\tau_{u,p},\nu_{u,p})\in\Gamma_u\cap\Lambda_{n_{\mathrm{an}},n_{\mathrm{de}},n_{\mathrm{do}}}}\beta_{u,p}^2$.

III. Channel Estimation and Pilot Design

A. Optimal Channel Estimation for Pilot Segment

We consider the estimation of the SFT domain channel at pilot segments. According to (8) and the frame structure shown in Fig. 1, the SFT domain channel at pilot segments is

$$\mathbf{h}_u^{\mathrm{SFT,p}} = \tilde{\mathbf{P}}\mathbf{h}_u^{\mathrm{TB}}\in\mathbb{C}^{MN_{\mathrm{v}}N_{\mathrm{F}}\times 1}, \quad (9)$$

where $\tilde{\mathbf{P}}\triangleq(\mathbf{\Theta}\otimes\mathbf{I}_{N_{\mathrm{v}}}\otimes\mathbf{I}_M)\mathbf{P}$, $\mathbf{\Theta}\in\mathbb{C}^{N_{\mathrm{F}}\times N}$ with the n_{F}-th row being the $(n_{\mathrm{F}}N_{\mathrm{S}}+n_{\mathrm{p}})$-th row of $\mathbf{I}_N$, and $\mathbf{I}_N$ is the $N\times N$ identity matrix.

In each timeslot in Fig. 1, the channel estimation should be performed to obtain the CSI using the pilot. We combine the current timeslot with the previous $N_{\mathrm{F}}-1$ timeslots to form a complete frame as in Fig. 1, where the current timeslot refers to the last timeslot in the whole frame.

Let $\mathbf{x}_u^{\mathrm{p}}\in\mathbb{C}^{N_{\mathrm{v}}\times 1}$ denote the transmitted pilot of UT u at valid subcarriers. The received signal can be given by

$$\mathbf{y}=\sum_{u=0}^{U-1}\mathbf{X}_u\mathbf{h}_u^{\mathrm{SFT,p}}+\mathbf{z}=\mathbf{X}\mathbf{h}^{\mathrm{SFT,p}}+\mathbf{z}=\mathbf{A}\mathbf{h}^{\mathrm{TB}}+\mathbf{z}, \quad (10)$$

where $\mathbf{h}^{\mathrm{SFT,p}}\triangleq\left[\left(\mathbf{h}_0^{\mathrm{SFT,p}}\right)^{\mathrm{T}},\left(\mathbf{h}_1^{\mathrm{SFT,p}}\right)^{\mathrm{T}},\cdots,\left(\mathbf{h}_{U-1}^{\mathrm{SFT,p}}\right)^{\mathrm{T}}\right]^{\mathrm{T}}$, $\mathbf{X}_u\triangleq\mathbf{I}_{N_{\mathrm{F}}}\otimes(\mathrm{diag}\{\mathbf{x}_u^{\mathrm{p}}\}\otimes\mathbf{I}_M)$, $\mathbf{X}\triangleq[\mathbf{X}_0,\mathbf{X}_1,\cdots,\mathbf{X}_{U-1}]$, $\mathbf{h}^{\mathrm{TB}}\triangleq\left[\left(\mathbf{h}_0^{\mathrm{TB}}\right)^{\mathrm{T}},\left(\mathbf{h}_1^{\mathrm{TB}}\right)^{\mathrm{T}},\cdots,\left(\mathbf{h}_{U-1}^{\mathrm{TB}}\right)^{\mathrm{T}}\right]^{\mathrm{T}}$, $\mathbf{A}\triangleq\mathbf{X}\left(\mathbf{I}_U\otimes\tilde{\mathbf{P}}\right)$, and $\mathbf{z}$ is the additive white Gaussian noise with distribution $\mathcal{CN}\left(\mathbf{z};\mathbf{0},\sigma_{\mathrm{z}}^2\mathbf{I}_{MN_{\mathrm{v}}N_{\mathrm{F}}}\right)$. Since the TB domain channel $\mathbf{h}^{\mathrm{TB}}$ is distributed as the complex Gaussian distribution, the optimal estimate of $\mathbf{h}^{\mathrm{TB}}$ is the MMSE estimate, which is given by

$$\hat{\mathbf{h}}^{\mathrm{TB}}=\mathbf{R}^{\mathrm{TB}}\mathbf{A}^{\mathrm{H}}\left(\mathbf{A}\mathbf{R}^{\mathrm{TB}}\mathbf{A}^{\mathrm{H}}+\sigma_{\mathrm{z}}^2\mathbf{I}_{MN_{\mathrm{v}}N_{\mathrm{F}}}\right)^{-1}\mathbf{y}, \quad (11)$$

where $\mathbf{R}^{\mathrm{TB}}=\mathbb{E}\left\{\mathbf{h}^{\mathrm{TB}}\left(\mathbf{h}^{\mathrm{TB}}\right)^{\mathrm{H}}\right\}=\mathrm{diag}\left\{\mathbf{R}_0^{\mathrm{TB}},\mathbf{R}_1^{\mathrm{TB}},\cdots,\mathbf{R}_{U-1}^{\mathrm{TB}}\right\}$. According to (9), the estimate of $\mathbf{h}^{\mathrm{SFT,p}}$ is expressed as

$$\hat{\mathbf{h}}^{\mathrm{SFT,p}}=\mathbf{R}^{\mathrm{SFT,p}}\mathbf{X}^{\mathrm{H}}\left(\mathbf{X}\mathbf{R}^{\mathrm{SFT,p}}\mathbf{X}^{\mathrm{H}}+\sigma_{\mathrm{z}}^2\mathbf{I}_{MN_{\mathrm{v}}N_{\mathrm{F}}}\right)^{-1}\mathbf{y}, \quad (12)$$

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where

$$\mathbf{R}^{\mathrm{SFT,p}}=\left(\mathbf{I}_U\otimes\tilde{\mathbf{P}}\right)\mathbf{R}^{\mathrm{TB}}\left(\mathbf{I}_U\otimes\tilde{\mathbf{P}}\right)^{\mathrm{H}}=\mathbb{E}\left\{\mathbf{h}^{\mathrm{SFT,p}}\left(\mathbf{h}^{\mathrm{SFT,p}}\right)^{\mathrm{H}}\right\}$$
$$=\mathrm{diag}\left\{\mathbf{R}_0^{\mathrm{SFT,p}},\mathbf{R}_1^{\mathrm{SFT,p}},\cdots,\mathbf{R}_{U-1}^{\mathrm{SFT,p}}\right\},\quad(13)$$

Note that (12) is also the MMSE estimate of $\mathbf{h}^{\mathrm{SFT,p}}$.

As an important measure for channel estimation, the NMSE of $\hat{\mathbf{h}}^{\mathrm{SFT,p}}$ is given by

$$\mathrm{NMSE}\triangleq\mathbb{E}\left\{\left\|\mathbf{h}^{\mathrm{SFT,p}}-\hat{\mathbf{h}}^{\mathrm{SFT,p}}\right\|^2\right\}\Big/\mathbb{E}\left\{\left\|\mathbf{h}^{\mathrm{SFT,p}}\right\|^2\right\}$$
$$=\sum_{u=0}^{U-1}\frac{1}{MN_{\mathrm{v}}N_{\mathrm{F}}U\vartheta_u}\mathrm{tr}\{\mathbf{R}_u^{\mathrm{SFT,p}}-\mathbf{R}_u^{\mathrm{SFT,p}}\mathbf{X}_u^{\mathrm{H}}\mathbf{C}^{-1}\mathbf{X}_u\mathbf{R}_u^{\mathrm{SFT,p}}\},$$

where $\mathbf{C}\triangleq\sum_{u=0}^{U-1}\mathbf{X}_u\mathbf{R}_u^{\mathrm{SFT,p}}\mathbf{X}_u^{\mathrm{H}}+\sigma_{\mathrm{z}}^2\mathbf{I}_{MN_{\mathrm{v}}N_{\mathrm{F}}}$, and $\vartheta_u\triangleq\sum_{p=0}^{P_u-1}\beta_{u,p}^2$ is the large-scale fading between the BS and UT u. For UT u, the term $\sum_{u=0}^{U-1}\mathbf{X}_u\mathbf{R}_u^{\mathrm{SFT,p}}\mathbf{X}_u^{\mathrm{H}}$ in $\mathbf{C}$ implies that the NMSE of each UT is related to not only its own statistical CSI but also that of other UTs, which is referred to as the inter-UT interference.

B. Pilot Design

We next discuss the pilot design for the optimal channel estimation. Motivated by the widely used phase shift pilots [16], the transmitted pilot of UT u is given by

$$\mathbf{x}_u^{\mathrm{P}}=\sigma_{\mathrm{p}}\mathbf{x}_{\mathrm{c}}\circ\left[e^{-\bar{\jmath}2\pi k_0\frac{N_\tau\phi_u}{N_{\mathrm{de}}N_{\mathrm{v}}}},\cdots,e^{-\bar{\jmath}2\pi k_{N_{\mathrm{v}}-1}\frac{N_\tau\phi_u}{N_{\mathrm{de}}N_{\mathrm{v}}}}\right]^{\mathrm{T}},\quad(14)$$

where σ_{p} is the square root of the pilot transmit power, $\phi_u\in\{0,N_{\mathrm{de}},\cdots,(\lfloor N_{\mathrm{v}}/N_\tau\rfloor-1)N_{\mathrm{de}}\}$ is referred to as the phase shift factor and $\mathbf{x}_{\mathrm{c}}$ is a sequence with unity modulus elements. A good choice for $\mathbf{x}_{\mathrm{c}}$ is the Zadoff-Chu (ZC) sequence, which has been used in the fifth generation (5G) wireless networks.

Next, we properly allocate all pilot sequences to each UT and optimize the channel estimation performance. To this end, we have the following theorem, which provides the criterion for the UT grouping and pilot scheduling, and its proof can be found in [17].

Theorem 1: When $M,N_{\mathrm{v}},N_{\mathrm{F}}\to\infty$, the minimum value of NMSE is given by

$$\mathrm{NMSE}_{\min}=\sum_{u=0}^{U-1}\frac{1}{MN_{\mathrm{v}}N_{\mathrm{F}}U\vartheta_u}\mathrm{tr}\Big\{\mathbf{R}_u^{\mathrm{SFT,p}}$$
$$-\mathbf{R}_u^{\mathrm{SFT,p}}\left(\mathbf{R}_u^{\mathrm{SFT,p}}+\frac{\sigma_{\mathrm{z}}^2}{\sigma_{\mathrm{p}}^2}\mathbf{I}_{MN_{\mathrm{v}}N_{\mathrm{F}}}\right)^{-1}\mathbf{R}_u^{\mathrm{SFT,p}}\Big\},\quad(15)$$

provided that, for $\forall u,u'$ and $u\neq u'$, one of the following conditions is satisfied

1) $\phi_u\neq\phi_{u'}$, $\mathbf{R}_u^{\mathrm{TB}}\mathbf{R}_{u'}^{\mathrm{TB}}\neq\mathbf{0}$,
2) $\mathbf{R}_u^{\mathrm{TB}}\mathbf{R}_{u'}^{\mathrm{TB}}=\mathbf{0}$.

From Theorem 1, when the NMSE achieves the minimum value, the NMSE of the channel estimate of each UT is only related to its own statistical CSI and is not related to other UTs, which means that the inter-UT interference is eliminated. Furthermore, condition 1) implies that to minimize the NMSE, UTs with overlapping TB domain channels should be allocated pilot sequences with different phase shift factors while condition 2) implies that UTs with non-overlapping TB domain channels can reuse the same pilot sequence.

Next, we define the degree of channel overlap between UTs u and u' as

$$\rho_{u,u'}\triangleq\frac{\mathrm{tr}\left\{\mathbf{R}_u^{\mathrm{TB}}\mathbf{R}_{u'}^{\mathrm{TB}}\right\}}{\left\|\mathbf{R}_u^{\mathrm{TB}}\right\|_{\mathrm{F}}\left\|\mathbf{R}_{u'}^{\mathrm{TB}}\right\|_{\mathrm{F}}}.\quad(16)$$

It can be directly checked that $0\leq\rho_{u,u'}\leq1$. In practical systems, the conditions in Theorem 1 are hard to be well satisfied due to limited M, N_{v}, and N_{F}. However, the power of the TB domain channel of each UT is typically concentrated on a narrow support and the channel overlap degree among UTs with different supports of TB domain channel can be small. Therefore, we can divide all UTs into S groups ($S\leq\lfloor N_{\mathrm{v}}/N_\tau\rfloor$) according to following UT grouping criterion.

i). The degree of channel overlap between any two UTs in the same group should be as low as possible.
ii). UTs with a high degree of channel overlap should be allocated to different groups.

From the criterion, a dedicated UT grouping algorithm can be developed [17]. Meanwhile, we allocate pilot sequences with different phase shift factors to each UT group. With UT grouping and pilot scheduling, the inter-UT interference is suppressed and the channel estimation performance is improved.

Note that the estimated SFT domain channel in the above discussion is at the pilot segment. When the channel is fast-varying and the channel at the pilot segment cannot be directly applied to the data segment, we can readily predict the channels at the data segment based on the estimated TB domain channel according to (8).

IV. CBFEM Based Channel Estimation

Although the MMSE estimation can achieve the optimal channel estimation performance, its complexity is unaffordable in practical systems. On the other hand, in HF skywave massive MIMO-OFDM communications, the limited propagation path number, Doppler spread, and small angle spread lead to the sparsity of the TB domain channel. Therefore, the channel estimation can be formulated as a sparse signal recovery problem and solved within the optimization framework of CBFEM, which has a clear objective function and can derive different solving algorithms through different constraints. Hence, we propose to use the CBFEM technique to estimate $\mathbf{h}^{\mathrm{TB}}$.

From (10), we have *a posterior* probability density $p\left(\mathbf{h}^{\mathrm{TB}},\mathbf{w}\,|\mathbf{y}\right)$ as follows

$$p\left(\mathbf{h}^{\mathrm{TB}},\mathbf{w}\,|\mathbf{y}\right)$$
$$=\frac{1}{p\left(\mathbf{y}\right)}\prod_{i=0}^{MN_{\mathrm{v}}N_{\mathrm{F}}-1}p\left(y_i\,|w_i\right)\prod_{i=0}^{MN_{\mathrm{v}}N_{\mathrm{F}}-1}p\left(w_i\,|\mathbf{h}^{\mathrm{TB}}\right)\prod_{j=0}^{N_{\mathrm{an}}N_{\mathrm{de}}N_{\mathrm{do}}U-1}p\left(h_j^{\mathrm{TB}}\right),$$

where $\mathbf{w}=\mathbf{A}\mathbf{h}^{\mathrm{TB}}\in\mathbb{C}^{MN_{\mathrm{v}}N_{\mathrm{F}}\times1}$ is the auxiliary vector, $p\left(y_i\,|w_i\right)=\mathcal{CN}\left(y_i;w_i,\sigma_{\mathrm{z}}^2\right)$, $p\left(w_i\,|\mathbf{h}^{\mathrm{TB}}\right)=\delta\left(w_i-\mathbf{a}_i\mathbf{h}^{\mathrm{TB}}\right)$, $p\left(h_j^{\mathrm{TB}}\right)=\mathcal{CN}\left(h_j^{\mathrm{TB}};0,\left[\mathbf{R}^{\mathrm{TB}}\right]_{j,j}\right)$, $\mathbf{a}_i$ is

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the i-th row of $\mathbf{A}$, h_j^{TB} is the j-th element of $\mathbf{h}^{\mathrm{TB}}$, and y_i and w_i are the i-th element of $\mathbf{y}$ and $\mathbf{w}$, respectively. Denote $b_{y,i}(w_i)$, $b_{w,i}\left(w_i, \mathbf{h}^{\mathrm{TB}}\right)$, and $b_{h,j}\left(h_j^{\mathrm{TB}}\right)$ as the factor beliefs of $p(y_i|w_i)$, $p\left(w_i\left|\mathbf{h}^{\mathrm{TB}}\right.\right)$, and $p\left(h_j^{\mathrm{TB}}\right)$, respectively. Let $q_{w,i}(w_i)$ and $q_{h,j}\left(h_j^{\mathrm{TB}}\right)$ denote the variable beliefs of w_i and h_j^{TB}, respectively. Then we can obtain the Bethe free energy as follows [9]

$$F_{\mathrm{B}}(b) = \sum_{i=0}^{MN_{\mathrm{v}}N_{\mathrm{F}}-1} \mathbb{D}\left\{b_{y,i}\left\|p(y_i|w_i)\right.\right\} + \sum_{i=0}^{MN_{\mathrm{v}}N_{\mathrm{F}}-1} \mathbb{D}\left\{b_{w,i}\left\|p\left(w_i\left|\mathbf{h}^{\mathrm{TB}}\right.\right)\right.\right\}$$
$$+ \sum_{j=0}^{N_{\mathrm{an}}N_{\mathrm{de}}N_{\mathrm{do}}U-1} \mathbb{D}\left\{b_{h,j}\left\|p\left(h_j^{\mathrm{TB}}\right)\right.\right\} + \mathbb{H}\left\{q_{w,i}\right\} + \sum_{j=0}^{N_{\mathrm{an}}N_{\mathrm{de}}N_{\mathrm{do}}U-1} MN_{\mathrm{v}}N_{\mathrm{F}}\mathbb{H}\left\{q_{h,j}\right\},$$

where $\mathbb{H}\{\cdot\}$ and $\mathbb{D}\{\cdot\}$ denote the entropy and relative entropy, respectively. Moreover, the beliefs need to fulfill the marginalization consistency constraints, which can be relaxed into the first-order and second-order moment consistency constraints as follows

$$\mathbb{E}\{w_i|b_{y,i}\} = \mathbb{E}\{w_i|b_{w,i}\} = \mathbb{E}\{w_i|q_{w,i}\} \tag{17a}$$

$$\mathbb{E}\left\{h_j^{\mathrm{TB}}|b_{h,j}\right\} = \mathbb{E}\left\{h_j^{\mathrm{TB}}|b_{w,i}\right\} = \mathbb{E}\left\{h_j^{\mathrm{TB}}|q_{h,j}\right\} \tag{17b}$$

$$\mathbb{E}\left\{|w_i|^2|b_{y,i}\right\} = \mathbb{E}\left\{|w_i|^2|b_{w,i}\right\} = \mathbb{E}\left\{|w_i|^2|q_{w,i}\right\} \tag{17c}$$

$$\mathbb{E}\left\{\left|h_j^{\mathrm{TB}}\right|^2|b_{h,j}\right\} = \frac{\sum_{i=0}^{MN_{\mathrm{v}}N_{\mathrm{F}}-1}\mathbb{E}\left\{\left|h_j^{\mathrm{TB}}\right|^2|b_{w,i}\right\}}{MN_{\mathrm{v}}N_{\mathrm{F}}} = \mathbb{E}\left\{\left|h_j^{\mathrm{TB}}\right|^2|q_{h,j}\right\} \tag{17d}$$

where (17d) is because $b_{w,i}$ contains a variable h_j^{TB} for any i.

Finally, we transform the channel estimation into the following CBFEM question

$$\min_{\substack{\{b_{y,i}\},\{b_{w,i}\},\{b_{h,j}\},\\ \{q_{w,i}\},\{q_{h,j}\}}} F_{\mathrm{B}}(b) \quad \text{s.t.}(17). \tag{18}$$

We can utilize Lagrange multipliers to solve the CBFEM question, and the detailed process can be found in [17]. The resulting CBFEM based channel estimation algorithm is summarized in Algorithm 2, where $b_h \triangleq \prod_{j=0}^{N_{\mathrm{an}}N_{\mathrm{de}}N_{\mathrm{do}}U-1} b_{h,j}$, $\mathbf{e} \triangleq [1, 1, \cdots, 1]^{\mathrm{T}} \in \mathbb{C}^{N_{\mathrm{an}}N_{\mathrm{de}}N_{\mathrm{do}}U\times 1}$, $\mathrm{Var}\{\cdot\}$ denotes the variance, and $(\cdot)^{\circ -1}$ denotes the element-wise inverse. The proposed algorithm has low complexity since the complexity of each iteration is only $\mathcal{O}\left(MN_{\mathrm{v}}N_{\mathrm{F}}N_{\mathrm{ave}}^{\mathrm{TB}}U\right)$, where $N_{\mathrm{ave}}^{\mathrm{TB}}$ is the average number of non-zero elements of $\mathbf{h}_u^{\mathrm{TB}}$ for each UT. Moreover, during the iterations, the damping factor can be leveraged to ensure the convergence of the algorithm [9].

Algorithm 1: CBFEM based channel estimation algorithm

Input: $\mathbf{A}$, $\mathbf{y}$, σ_{p}^2, σ_{z}^2, $p\left(\mathbf{h}^{\mathrm{TB}}\right) = \mathcal{CN}\left(\mathbf{h}^{\mathrm{TB}}; \mathbf{0}, \mathbf{R}^{\mathrm{TB}}\right)$

Initialize: $b_h = p\left(\mathbf{h}^{\mathrm{TB}}\right), \boldsymbol{\eta}^{h,b_h} = \mathbf{0}$

repeat

$\boldsymbol{\eta}^{h,b_w} = -\mathrm{Var}\left\{\mathbf{h}^{\mathrm{TB}}|b_h\right\}^{\circ -1} - \frac{1}{MN_{\mathrm{v}}N_{\mathrm{F}}}\boldsymbol{\eta}^{h,b_h}$

$\boldsymbol{\eta}^{h,b_h} = MN_{\mathrm{v}}N_{\mathrm{F}}\left(\left(\sigma_{\mathrm{p}}^2\mathbf{e}^{\mathrm{T}}\left(\boldsymbol{\eta}^{h,b_w}\right)^{\circ -1} - \sigma_{\mathrm{z}}^2\right)\sigma_{\mathrm{p}}^{-2}\mathbf{e} - \left(\boldsymbol{\eta}^{h,b_w}\right)^{\circ -1}\right)^{\circ -1}$

$\tilde{\boldsymbol{\tau}}^{h,b_w} = \mathbb{E}\left\{\mathbf{h}^{\mathrm{TB}}|b_h\right\} \circ \mathrm{Var}\left\{\mathbf{h}^{\mathrm{TB}}|b_h\right\}^{\circ -1}$

$\boldsymbol{\kappa} = \tilde{\boldsymbol{\tau}}^{h,b_w} \circ \left(\boldsymbol{\eta}^{h,b_w}\right)^{\circ -1}$

$\boldsymbol{\psi} = \mathbf{A}\boldsymbol{\kappa}$

$\boldsymbol{\varpi} = \frac{1}{MN_{\mathrm{v}}N_{\mathrm{F}}\sigma_{\mathrm{p}}^2}\mathbf{A}^{\mathrm{H}}(\mathbf{y} + \boldsymbol{\psi}) - \boldsymbol{\kappa}$

$b_h \propto p\left(\mathbf{h}^{\mathrm{TB}}\right)\mathcal{CN}\left(\mathbf{h}^{\mathrm{TB}}; \boldsymbol{\varpi}, \mathrm{diag}\left\{-\left(\boldsymbol{\eta}^{h,b_h}\right)^{\circ -1}\right\}\right)$

until the termination condition is fulfilled;

Output: $\hat{\mathbf{h}}^{\mathrm{TB}} = \mathbb{E}\left\{\mathbf{h}^{\mathrm{TB}}|b_h\right\}$

V. Simulation Results

In this section, we provide simulation results to illustrate the performance of the proposed channel estimation approach for HF skywave massive MIMO-OFDM communications. The main simulation parameters are given in Table I. We generate UTs at a distance of 2000 km from the BS. A commercial ray-tracing software, Proplab-Pro version 3.1, is utilized to generate the realistic HF skywave channel for each UT [18]. Specifically, we use the software to get parameters of each propagation path between the BS and each UT, including the path gain, the propagation delay, the azimuth and elevation AoA. The maximum Doppler comes from the ionosphere and UT mobility, i.e., $\nu_{\max} = \frac{\nu_{\mathrm{iono}}}{2} + \frac{v_u}{c}f_c$, where ν_{iono} is the Doppler spread imposed by the ionosphere and v_u is the velocity of UT u. In the simulation, ν_{iono} is set to be 0.5 Hz for moderate ionospheric conditions at mid-latitude regions [19] and v_u is set to be 100 km/h, the Doppler frequency of each path is randomly generated within the range $[-\nu_{\max}, \nu_{\max})$, and N_{d} is set to be 8 according to the maximum Doppler. Then, the SFT domain channel of each UT is generated according to (5) and the statistical CSI $\mathbf{R}^{\mathrm{TB}}$ is acquired by the method proposed in [20].

The signal-to-noise ratio (SNR) in the simulation refers to the received SNR. The UT grouping according to the criterion in Section III-B is referred to as TB-UG, and the number of groups to divide is set to be $S = \lfloor N_{\mathrm{v}}/N_\tau \rfloor$. The performance of random UT grouping (Random-UG) is also provided for comparison, and we refer to the CBFEM based channel estimation algorithm as the CBFEM-CE algorithm.

TABLE I
Simulation Parameters

Parameter	Value	Parameter	value
Carrier frequency f_{c}	16 MHz	CP length N_{g}	512
Subcarrier spacing Δf	250 Hz	BS antenna spacing d	9 m
Subcarrier number N_{c}	2048	Frame size $(N_{\mathrm{F}}, N_{\mathrm{S}}, n_{\mathrm{p}})$	(8, 14, 6)

We first compare the NMSE performance of the MMSE estimation and the proposed CBFEM-CE algorithm with different fine factors. Due to the high complexity of the MMSE estimation, we use a set of smaller parameters in this simulation. Specifically, we set $M = 64$, $N_{\mathrm{v}} = 128$, $U = 32$. The simulation results are shown in Fig. 2, where the UT grouping method is TB-UG. From the figure, larger fine factors can achieve better performance due to the improved accuracy

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of the proposed TB based channel model. Furthermore, the performance of the CBFEM-CE algorithm can approach that of the optimal MMSE estimation with a very small gap and low complexity.

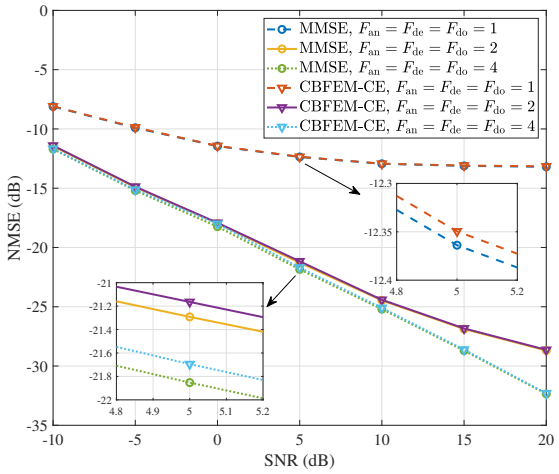


Fig. 2. The NMSE performance of the MMSE estimation and the CBFEM-CE algorithm with different fine factors.

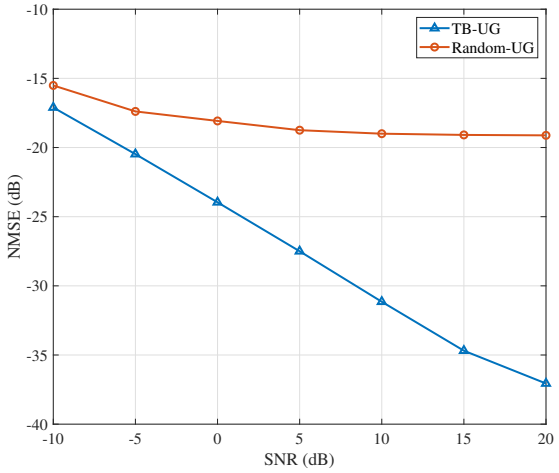


Fig. 3. The NMSE performance of different UT grouping methods.

Fig. 3 compares the NMSE performance of different UT grouping methods, where $M = 128$, $N_v = 1536$, $U = 64$, $F_{an} = F_{de} = F_{do} = 2$, and the channel estimation algorithm is CBFEM-CE. From the figure, the proposed UT grouping method is crucial to the channel estimation and has a significant performance gain over random-UG, especially when the SNR is high. This is because there is severe inter-UT interference when the number of UTs is large, which can be effectively tackled by the proposed UT grouping method.

VI. Conclusion

In this paper, we have investigated the channel estimation for HF skywave massive MIMO-OFDM communications. We first established a TB based channel model using sampled triple steering vectors. Then, based on the channel model, we investigated the optimal channel estimation for pilot segments and revealed the conditions for minimizing the NMSE of the channel estimate. To reduce the complexity of the channel estimation, we developed the CBFEM based channel estimation algorithm. Simulation results verified the validity of the proposed CSI acquisition approach.

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